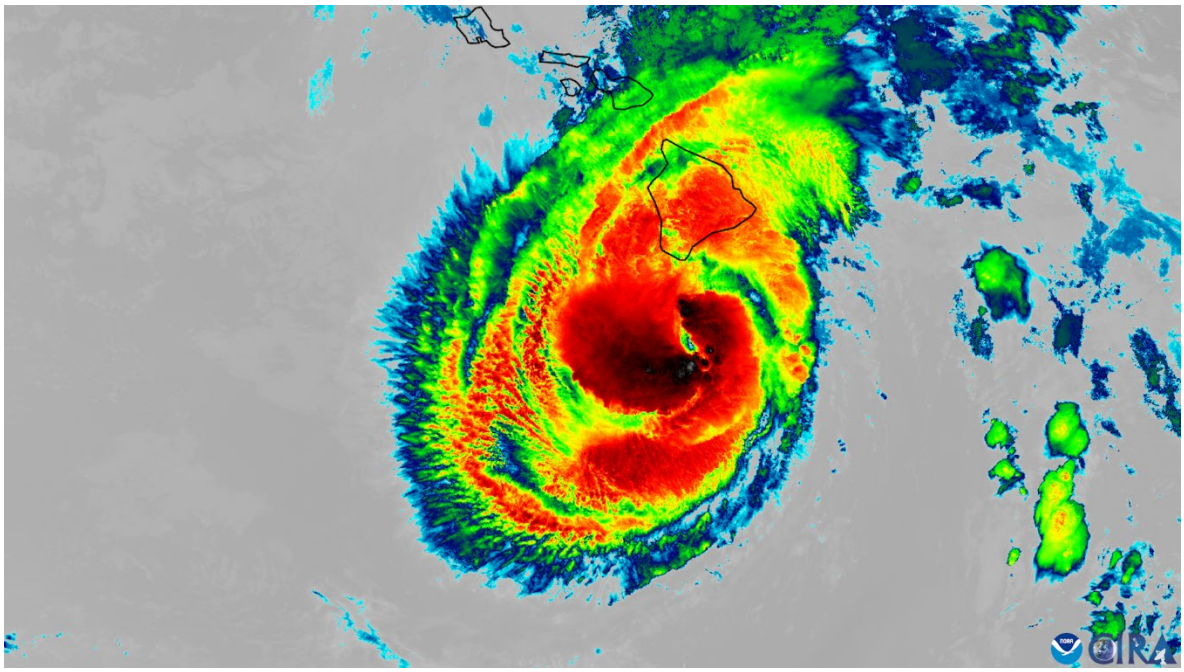
CENTRAL PACIFIC HURRICANE CENTER TROPICAL CYCLONE REPORT

HURRICANE HONE

(CP012024)

22 August–1 September 2024

Alex Gibbs
Jon Jelsema
Central Pacific Hurricane Center
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VIIRS LONGWAVE IR WINDOW (BAND 15) SATELLITE IMAGE OF HURRICANE HONE JUST SOUTH OF THE BIG ISLAND OF HAWAII AT 1149 UTC 25 AUGUST 2024. IMAGE COURTESY OF: NOAA/CIRA/CSU

Hurricane Hone was a category 1 hurricane (on the Saffir-Simpson Hurricane Wind Scale) that brought heavy rainfall and gusty winds to portions of the Hawaiian Islands as it passed just south of the Big Island of Hawaii.

Hurricane Hone

22 AUGUST – 1 SEPTEMBER 2024

SYNOPTIC HISTORY

The origin of Hone can be traced back to a disturbance that formed within a broad and convectively active monsoon trough around 1500 n mi to the east-southeast of the Hawaiian Islands over the western portion of the eastern Pacific Ocean on 18 August. Showers and thunderstorms associated with the disturbance gradually organized during the next several days, while the system moved slowly westward. The disturbance that Hone would eventual develop from was also interacting with another disturbance to its southwest near 10ºN 140°W, which appears to have contributed to its slow development. The dominant disturbance to the east, developed a well-defined circulation and sufficient convection to be designated a tropical depression by 0000 UTC 22 August, when it was located about 1000 n mi east-southeast of the Hawaiian Islands. The depression crossed into the Central Pacific basin around 1200 UTC 22 August, and it strengthened into Tropical Storm Hone by 1800 UTC that day[1]. The "best track" chart of Hone's path is given in Fig. 1, with the wind and pressure histories shown in Figs. 2 and 3, respectively. The best track positions and intensities are listed in Table 1[2].

Hone moved steadily west-northwestward across the Central Pacific during the next several days along the southern periphery of a large subtropical ridge to the north. The storm gradually strengthened on its approach to the Hawaiian Islands, fighting off periodic intrusions of drier mid-level air, while otherwise in a conducive environment for strengthening influenced by low vertical wind shear and warm sea surface temperatures of 27-–28°C. Observations from the 53rd Weather Reconnaissance Squadron of the U.S. Air Force Reserve Command, provided valuable data on the cyclone's structure and intensity as it neared Hawaii.

By 0600 UTC 25 August, while located about 100 n mi south of Hilo on the Big Island of Hawaii, satellite, aircraft and radar data indicated that Hone had reached hurricane strength, with maximum sustained winds estimated to be 65 kt. Hone strengthened further while making its closest approach to the Hawaiian Islands, reaching a peak intensity of 75 kt by 1200 UTC 25 August. At that time, Hone was centered about 45 n mi south of South Point on the Big Island. Hone brought tropical–storm–force winds and heavy rainfall, particularly over windward and southeast-facing slopes of the Big Island, as its center passed just south of the island.

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[1] Note that genesis was operationally determined to have occurred in the central Pacific basin, however, after post-storm analysis it was determined to have occurred 12 hours earlier while the cyclone was in the eastern Pacific basin. As a result, the storm was given a central Pacific (CP) storm identifier in real-time as opposed to an eastern Pacific (EP) storm identifier.

[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

Hone moved steadily westward, to the south of the Hawaiian Island chain for the next couple of days, steered along the southern periphery of a subtropical ridge to the north of Hawaii. The cyclone weakened into a tropical storm around 0000 UTC 26 August, as increasing westerly vertical wind shear and entrainment of drier mid-level took its toll on the cyclone. Hone maintained tropical storm strength for the next several days while it continued to track westward, with the cyclone gradually weakening due to persistent vertical wind shear, causing deep convection to become increasingly intermittent.

Hone weakened into a tropical depression at 0000 UTC 30 August over the western portion of the Central Pacific basin, around 750 n mi west of Kauai. Convection periodically pulsed over the cyclone during the next day or so. After 0000 UTC 31 August, vertical wind shear decreased and cooler temperatures aloft associated with an upper-level low caused deep convection to increase over the low-level circulation. Hone regained tropical storm status by 0600 UTC 31 August, with the cyclone remaining a tropical storm for another 24 hours or so. During that time, Hone slowed down and turned toward the north, steered between the upper-level low to the northwest and a subtropical ridge to the east. Deep convection waned, and Hone degenerated to a post-tropical low by 0600 UTC 1 September and dissipated later that day as the low-level circulation became absorbed into a larger circulation associated with the nearby upper-level low.

METEOROLOGICAL STATISTICS

Observations in Hone include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB), the Satellite Analysis Branch (SAB), the Central Pacific Hurricane Center (PHFO), and the Joint Typhoon Warning Center (JTWC). Objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates were from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Observations also include flight-level and dropwindsonde observations from 5 flights (Fig. 4) of the 53rd Weather Reconnaissance Squadron of the U.S. Air Force Reserve Command (AFRES). Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Hone.

Buoy observations of winds of tropical storm force associated with Hone are given in Table 2, and selected surface observations from land stations are given in Table 3. There were no ship reports of tropical-storm-force winds associated with Hone.

Winds and Pressure

Hone's estimated peak intensity of 75 kt occurred around 1200 UTC 25 August, as it passed by approximately 45 n mi south of South Point, Hawaii, as a Category 1 hurricane on the Saffir-Simpson Hurricane Wind Scale. This peak intensity is based on maximum 700 mb flight-level winds of 86 kt measured in the northeast eyewall during the 4th reconnaissance mission

flown by the U.S. Air Force 53rd Weather Reconnaissance Squadron at 1113 UTC 25 August, approximately 20 n mi from the center. This flight level wind corresponds to 77 kt surface winds using standard reduction factors. Doppler radar from the Naalehu, Hawaii WSR-88D corroborated these strong winds, estimating 92 kt at 6,500 ft around the same time, which supports 75 to 80 kt surface winds using standard reduction factors. A raw subjective Dvorak data T-number from PHFO/CPHC of T4.5/77 kt, also supported this peak intensity. The minimum central pressure at peak intensity was estimated to be 988 mb, based largely on aircraft dropsonde data received at 1123 UTC 25 August. The dropsonde measured a surface pressure of 991 mb with 6 kt of wind at the surface (10 m), which supports a minimum surface pressure of 990 mb. The dropsonde observation was taken during a period of intensification however, so the minimum pressure at peak intensity accounts for some slight lowering of the surface pressure between the time of the dropsonde observation and 1200 UTC that day. Note that no reconnaissance missions were conducted during the estimated peak intensity of Hone, with a gap in center fixes occurring between 1123 UTC and 1708 UTC 25 August.

Widespread wind gusts exceeding 50 kt were observed across the Big Island, with a peak gust of 63 kt recorded at Kohala Ranch. Several locations reported sustained winds of 35–44 kt. (Table 3). The center of Hone passed directly over National Data Buoy Center (NDBC) buoy 51003 around 1500 UTC 26 August. Prior to the center passing over the buoy, sustained winds of 43 kt were measured with a peak wind gust of 55 kt around 1440 UTC 26 August. It should be noted that the peak wind gust occurred in seas measuring over 22 feet.

Storm Surge[3]

A NOAA tide gauge at Hilo Bay (Kuhio Bay) on the eastern side of the Big Island recorded a peak water level of 0.73 ft (0.22 m) above Mean Higher High Water (MHHW) during Hone's closest approach on 25 August. This modest rise was likely caused by a combination of storm surge and wave setup from large breaking waves. No coastal inundation or damage was reported. Higher localized water levels may have occurred along the more exposed southeast-facing shores of the Big Island, particularly near South Point, due to increased wave setup and runup. However, no tide gauges or direct observations were available in that region.

Rainfall and Flooding

On 24–25 August as the center of Hone passed south of the Island of Hawaii (Big Island), rain bands associated with the tropical cyclone moved over the east- and southeast-facing slopes of the island, producing heavy rainfall and flash flooding in several areas. The maximum 2-day

[3] Several terms are used to describe water levels due to a storm. **Storm surge** is defined as the abnormal rise of water generated by a storm, over and above the predicted astronomical tide, and is expressed in terms of height above normal tide levels. Because storm surge represents the deviation from normal water levels, it is not referenced to a vertical datum. **Storm tide** is defined as the water level due to the combination of storm surge and the astronomical tide, and is expressed in terms of height above a vertical datum, i.e. the North American Vertical Datum of 1988 (NAVD88) or Mean Lower Low Water (MLLW). **Inundation** is the total water level that occurs on normally dry ground as a result of the storm tide, and is expressed in terms of height above ground level. At the coast, normally dry land is roughly defined as areas higher than the normal high tide line, or Mean Higher High Water (MHHW).

total was 33.58 inches at a CoCoRaHS gage at Volcano Village. Around half of the island received more than 10 inches, including a large area of 20+ inches on the eastern flank of the island upslope from the city of Hilo (Fig. 5). Other notable totals included 28.82 inches at the Hakalau Remote Automated Weather Station (RAWS), 25.03 inches at Saddle Quarry (USGS), 24.46 inches at Kulani NWR (NWS), and over 21 inches at both Mountain View and Waiākea Uka (Table 3).

Although accumulations were substantial, hourly rainfall rates generally remained below 1 inch per hour, which limited the severity of the flash flooding. However, numerous road closures occurred in low-lying areas, including Highway 11 at Kawa Flats, Cane Haul Road and Wood Valley Road near Pāhala, and North Kulani Road near Mountain View in the Puna District. Flooding damage was also reported to numerous structures across the Big Island, referenced in the Casualty and Damage Statistics section below.

CASUALTY AND DAMAGE STATISTICS

There were no confirmed fatalities or serious injuries associated with Hurricane Hone. Damage reports of downed trees and power lines were widespread, and power outages affected more than 24,000 customers statewide, with the most severe disruptions occurring on the Big Island of Hawaii. Numerous structures were also damaged due to flooding, with at least 38 damage claims submitted on the windward (eastern) side of the island from Kohala southward to Kau. One home, the former “Hobbit House” bed-and-breakfast in Waiohinu suffered extensive damage due to flash flooding (Fig. 6). At least 9 of the structures surveyed had major damage.

Additional wind impacts occurred on the islands of Maui and Oahu, which were associated with strong gradient winds between high pressure north of Hawaii and Hurricane Hone passing by to the south.

FORECAST AND WARNING CRITIQUE

The genesis timing of Hone was well anticipated. The disturbance from which Hone developed was first highlighted in the Tropical Weather Outlook (TWO) 144 h prior to formation (Table 4). The 7-day genesis probability was raised to the medium category (40-60%) 132 h before formation and to the high category (>60%) 114 h in advance. Two-day genesis forecasts also provided considerable lead time; low probabilities (<40%) were introduced 114 h prior to formation and high probabilities were introduced 36 h in advance of development. One unique genesis challenge leading up to the formation of Hone, was that two areas of potential genesis were highlighted near the border of the central and eastern Pacific basin around 140°W longitude. Although the genesis of Hone was well anticipated, identifying the specific disturbance from which the cyclone would develop was challenging. Due to this uncertainty, the initial genesis area had the 7-day probabilities lowered from the high category to the medium category 78 h prior to formation, and the 7-day and 2-day genesis probabilities were raised into the high category again 36 h prior to development (after the two systems being monitored for development merged into a single area of interest).

Forecasts of Hone's genesis location were highly accurate, with the system falling within each hatched-area genesis forecast, resulting in a 100% hit rate in NHC's Graphical TWOs (Fig.7).

A verification of CPHC official track forecasts for Hone is given in Table 5a. Official track forecast errors were lower than the mean official errors for the previous 5-yr period at forecast hours 12, 24 and 120, and greater than the mean official errors at all other forecast intervals. A homogeneous comparison of the official track errors with selected guidance models is given in Table 5b. The official track errors were lower than the errors of nearly every deterministic model at almost every time interval. In contrast, the consensus aids TVCE, TVCX, TVDG and HCCA outperformed the official forecast at most time intervals.

A verification of CPHC official intensity forecasts for Hone is given in Table 6a. Official intensity forecast errors were lower than the mean official errors for the previous 5-yr period at all time steps. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 6b. The official intensity errors outperformed nearly all guidance at almost every time interval, with very few exceptions.

It should be noted that the 5-yr mean track and intensity errors for the 60-h time interval were not available, as 60-h forecasts had not been produced for a consecutive 5-yr period, since they were introduced during the 2020 hurricane season.

Watches and warnings associated with Hone are given in Table 7.

IMPACT-BASED DECISION SUPPORT SERVICES & PUBLIC COMMUNICATION

The National Weather Service (NWS) Weather Forecast Office (WFO) in Honolulu and the Central Pacific Hurricane Center (CPHC) began providing impact-based decision support services (IDSS) to local, state, and federal emergency management partners as Hone approached the Hawaiian Islands. Daily coordination briefings and storm updates were conducted with Hawaii Emergency Management Agency (HI-EMA), county emergency managers, and other critical stakeholders beginning on 22 August and continuing through the duration of the storm's impact.

In addition, WFO Honolulu supported maritime partners and port officials, including the U.S. Coast Guard (USCG) District 14, with briefings and hazard updates related to strong winds, high surf, and heavy rainfall. Spot Forecasts were produced for the USCG as Hone approached the Hawaiian Islands, while the USCG Cutter *Kimball* was undergoing extensive dockside maintenance on Sand Island, Oahu, Hawaii. Based on CPHC forecasts, the USCG decided against having their contractors remove the tents covering the exposed deck of the vessel, saving them over $250,000. The office also coordinated closely with media outlets and the general public, issuing flood watches, high surf warnings, hurricane local statements, and continuous updates through web and social media platforms.

Frequent internal coordination with the CPHC and NWS Pacific Region Headquarters ensured a consistent messaging framework and supported continuity of operations across the forecast enterprise.

ACKNOWLEDGEMENTS

Dr. Lisa Bucci produced Figure 4. Dr. Philippe Papin produced Figure 6. Data in Table 3 were provided by the NWS WFO in Honolulu, Hawaii. Forecasters and the Hydrologist at WFO Honolulu contributed rainfall, surf, and wind data summaries. Input for the impact-based decision support section was also provided by WFO Honolulu.

Table 1. Best track for Hurricane Hone, 22 August–1 September, 2024.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
22 / 0000	15.5	137.5	1007	30	tropical depression
22 / 0600	15.4	138.8	1007	30	"
22 / 1200	15.4	140.2	1007	30	"
22 / 1800	15.5	141.5	1006	35	tropical storm
23 / 0000	15.7	142.8	1006	35	"
23 / 0600	15.9	144.1	1005	40	"
23 / 1200	16.3	145.6	1005	40	"
23 / 1800	16.6	147.1	1002	45	"
24 / 0000	16.9	148.5	1000	45	"
24 / 0600	17.2	149.9	999	50	"
24 / 1200	17.4	151.3	997	55	"
24 / 1800	17.6	152.6	1000	50	"
25 / 0000	17.9	153.8	1000	50	"
25 / 0600	18.1	154.8	991	65	hurricane
25 / 1200	18.2	155.7	988	75	"
25 / 1800	18.6	156.7	990	65	"
26 / 0000	19.0	157.9	994	60	tropical storm
26 / 0600	19.1	159.1	996	60	"
26 / 1200	19.1	160.1	998	55	"
26 / 1800	19.1	161.2	998	55	"
27 / 0000	19.4	162.3	1001	50	"
27 / 0600	19.7	163.5	1001	50	"
27 / 1200	19.8	164.7	1003	45	"
27 / 1800	20.0	165.6	1003	45	"
28 / 0000	20.2	166.5	1003	45	"
28 / 0600	20.3	167.3	1003	45	"
28 / 1200	20.4	168.1	1003	45	"
28 / 1800	20.5	168.9	1003	45	"
29 / 0000	20.6	169.7	1003	45	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
29 / 0600	20.8	170.6	1005	40	"
29 / 1200	21.0	171.5	1007	35	"
29 / 1800	21.3	172.2	1007	35	"
30 / 0000	21.4	173.0	1008	30	tropical depression
30 / 0600	21.5	173.7	1008	30	"
30 / 1200	21.7	174.4	1008	30	"
30 / 1800	21.8	175.3	1008	30	"
31 / 0000	21.9	175.7	1008	30	"
31 / 0600	22.1	175.7	1007	35	tropical storm
31 / 1200	22.3	175.9	1005	40	"
31 / 1800	22.5	176.3	1007	35	"
01 / 0000	23.3	176.5	1007	35	"
01 / 0600	23.8	176.5	1007	35	low
01 / 1200	-	-	-	-	dissipated
25 / 1200	18.2	155.7	988	75	maximum wind and minimum pressure

Table 2. Selected buoy observations with winds of at least 34 kt for Hurricane Hone, 22 August–1 September, 2024.

Date/Time (UTC)	**Buoy ID**	**Latitude (°N)**	**Longitude (°W)**	**Wind dir/ speed (kt)**	**Pressure (mb)**
26 / 1450	51003	19.2	160.6	010 / 39	1000.9
26 / 1510	51003	19.2	160.6	040 / 35	998.7

Table 3. Selected surface observations for Hurricane Hone, 22 August–1 September 2024.

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Total rain (in)
	Date/ time (UTC)	Press. (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)	
International Civil Aviation Organization (ICAO) Sites						
Bradshaw Army Airfield (PHSF) (19.76N 155.55W)			25/1055	30	44	
Hilo Intl. Aprt. (PHTO) (19.72N 155.06W)	25/1522	1012.0				
Keahole Aprt. Kona (PHKO) (19.74N 156.05W)	25/1405	1006.0				
Non-METAR Observations						
Helco Hokuloa Ucc (19.94N 155.67W)			25/1550	38	53	
Helco Hulukupuna St. (20.02N 155.81W)			25/1250	30	51	
Helco Kawaihae Rd. (20.03N 155.71W)			25/1300	39	56	
Helco Waikoloa Rd. 2 (19.93N 155.78W)			25/0600	36	51	
Kamuela/Waimea (PHMU) (20.00N 155.67W)			25/0448	44	49	
Kohala Ranch (20.09N 155.83W)			25/1400	35	63	
Puu Mali (19.92N 155.42W)			25/1000	35	51	
South Point (18.99N 155.67W)			25/0945	37	53	
Waikoloa (19.92N 155.79W)			25/0530	31	50	
Other						
Volcano Village (19.43N 155.24W)						33.58
Hakalau (19.82N 155.33W)						28.82
Honolii Stream (19.76N 155.15W)						19.63
Kawainui Stream (20.09N 155.68W)						18.16
Kulani NWR (19.55N 155.31W)						24.46

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Total rain (in)
	Date/ time (UTC)	Press. (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)	
Laupohoehoe (19.98N 155.23W)						9.78
Mountain View (19.57N 155.12W)						21.30
Nene Cabin (19.26N 155.61W)						18.08
Papaikou Well (19.79N 155.13W)						15.07
Puu Kukui (20.89N 156.59W)						10.44
Saddle Quarry (19.69N 155.29W)						25.03
Waikea Uka (19.66N 155.13W)						21.04
West Wailuaiki (20.81N 156.59W)						12.70

[a] Date/time is for sustained wind when both sustained and gust are listed.
[b] Except as noted, sustained wind averaging periods for C-MAN and land-based reports are 2 min; buoy averaging periods are 8 min.

Table 4. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	168-Hour Outlook
Low (<40%)	72	144
Medium (40%-60%)	(36)*	132
High (>60%)	36	114 (36)*

* Medium probabilities of formation were never introduced into the 2-day TWO. In the 7-day TWO, high probabilities for development were introduced a second time after being lowered into the medium category for several days.

Table 5a. CPHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Hone, 22 August–1 September 2024. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**20.3**	**31.3**	42.8	54.7	61.3	73.8	99.9	**97.8**
OCD5	24.6	42.6	54.1	62.0	67.3	71.0	90.9	106.1
Forecasts	37	35	33	31	29	27	18	13
OFCL (2019-23)	21.8	34.3	41.2	46.4	-	61.9	81.0	120.5
OCD5 (2019-23)	31.0	60.0	85.3	110.1	-	165.5	233.6	272.7

Table 5b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Hone, 22 August–1 September 2024. Errors smaller than the CPHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 5a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	18.1	26.6	38.8	52.0	58.9	72.2	86.9	55.1
OCD5	23.2	36.3	49.7	60.4	68.6	**71.8**	89.2	100.5
GFSI	20.7	32.0	50.1	71.7	87.5	102.2	104.1	101.9
HWFI	21.3	29.8	44.8	60.1	68.9	87.6	111.2	126.8
HMNI	22.7	32.9	40.9	55.5	73.7	110.3	210.1	316.7
HFAI	19.3	28.4	40.5	**48.2**	**55.6**	**71.5**	122.1	215.3
HFBI	20.7	32.3	49.8	71.0	88.5	111.7	114.9	110.9
EMXI	**16.3**	28.4	39.1	52.0	59.0	73.8	95.2	98.4
CMCI	24.0	39.3	53.9	66.3	72.7	86.7	92.0	82.7
CTCI	18.4	26.8	39.4	**50.5**	63.9	80.5	157.2	251.9
TVCE	**17.8**	**26.2**	**37.4**	**47.8**	**51.9**	**63.5**	**66.9**	82.0
TVCX	**17.6**	**25.8**	**37.7**	**47.5**	**51.1**	**62.1**	**68.0**	84.7
TVDG	**17.7**	**25.7**	39.0	**49.7**	**55.0**	**66.3**	**69.6**	79.2
GFEX	18.2	28.1	43.0	60.5	71.8	85.3	95.3	96.1
HCCA	18.6	28.0	**38.4**	**50.7**	**57.9**	**66.3**	**81.3**	108.1
FSSE	19.2	26.9	**38.6**	53.0	**58.8**	**67.4**	**77.8**	71.5
AEMI	18.1	30.8	47.1	66.0	80.2	94.8	105.5	102.6
NVGI	21.9	32.1	41.7	56.1	71.4	93.2	103.9	155.3
TABS	22.3	35.8	55.4	74.5	91.1	112.2	136.5	144.2
TABM	25.0	54.0	88.0	120.5	152.6	179.6	235.8	303.5
TABD	45.6	119.5	199.6	275.2	348.2	417.7	531.1	655.1
Forecasts	26	25	25	25	24	22	14	9

Table 6a. CPHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Hone, 22 August–1 September 2024. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**3.4**	**3.6**	**4.1**	**5.3**	5.0	**3.3**	**4.7**	**3.8**
OCD5	5.1	6.1	8.1	13.2	12.8	12.4	8.9	3.9
Forecasts	37	35	33	31	29	27	18	13
OFCL (2019-23)	4.5	8.7	8.7	10.3	-	11.6	16.3	16.5
OCD5 (2019-23)	6.2	9.4	14.5	19.0	-	25.1	17.5	21.9

Table 6b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Hone, 22 August–1 September 2024. Errors smaller than the CPHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 6a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	4.2	4.0	4.2	5.8	5.0	3.0	4.6	3.3
OCD5	4.6	6.3	8.0	12.8	12.5	11.2	8.2	4.1
HWFI	4.7	6.4	6.0	7.4	7.2	8.8	10.0	5.4
HMNI	5.7	7.3	6.4	7.8	5.9	8.0	9.2	11.9
HFAI	4.7	5.8	5.6	6.1	9.8	14.7	24.5	23.2
HFBI	5.5	6.6	8.9	11.7	12.6	15.4	12.9	7.0
DSHP	5.0	6.7	7.2	8.1	7.9	8.1	10.1	8.7
LGEM	4.8	6.6	8.2	8.8	8.4	6.1	5.8	6.8
ICON	4.7	5.9	5.6	5.9	5.1	4.1	**3.7**	6.7
IVCN	4.4	5.0	5.1	**5.2**	**4.2**	4.7	5.5	7.6
IVDR	4.4	4.9	5.0	**5.1**	**4.3**	5.5	6.2	7.7
CTCI	5.3	5.9	5.9	6.6	6.5	7.0	9.2	17.0
GFSI	5.2	7.4	8.2	9.2	9.4	8.7	11.9	13.3
EMXI	4.2	5.3	6.3	7.4	7.7	6.7	6.2	8.9
HCCA	4.6	4.6	5.3	5.9	5.3	7.7	10.3	5.4
FSSE	4.2	4.1	5.0	**5.7**	5.1	5.8	6.8	6.4
CMCI	4.7	7.0	8.1	9.2	9.9	9.3	11.8	11.4
AEMI	5.7	7.2	8.4	8.9	8.6	7.1	11.0	10.2
NVGI	4.3	5.9	6.2	6.9	6.0	5.7	9.2	10.0
Forecasts	26	25	25	25	24	22	13	9

Table 7. Watch and warning summary for Hurricane Hone, 22 August–1 September 2024.

Date/Time (UTC)	Action	Location
23/0300	Tropical Storm Watch issued	Big Island of Hawaii
24/0300	Tropical Storm Warning issued	Big Island of Hawaii
26/0000	Tropical Storm Warning discontinued	Big Island of Hawaii
30/2100	Tropical Storm Watch issued	Kure Atoll, Midway Atoll, and Papahanaumokuakea Marine National Monument from Lisianski Island to Pearl and Hermes Atoll
01/0900	Tropical Storm Watch discontinued	Papahanaumokuakea Marine National Monument from Lisianski Island to Pearl and Hermes Atoll
01/2100	Tropical Storm Watch discontinued	Kure Atoll and Midway Atoll

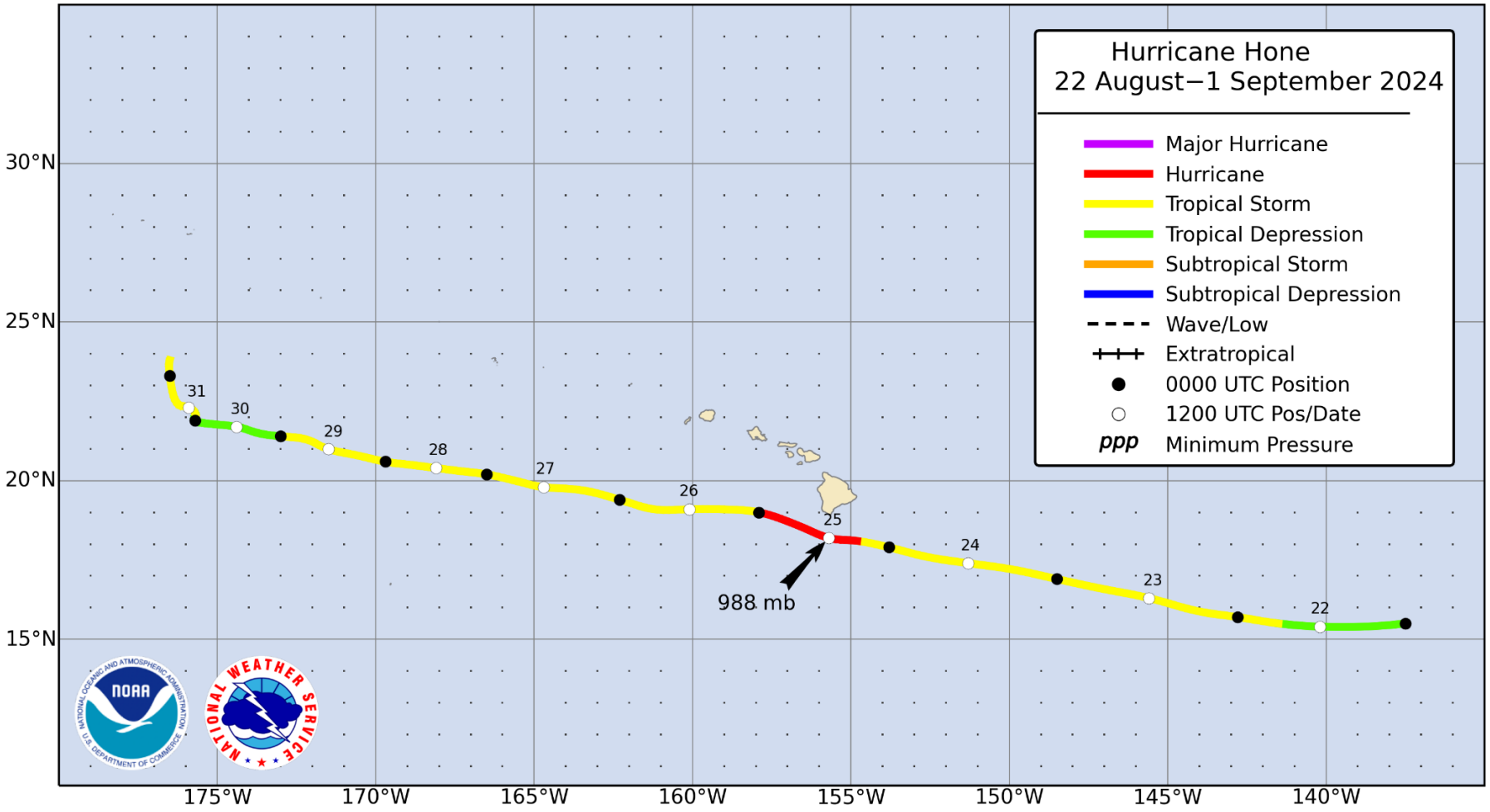


Figure 1. Best track positions for Hurricane Hone, 22 August–1 September 2024.

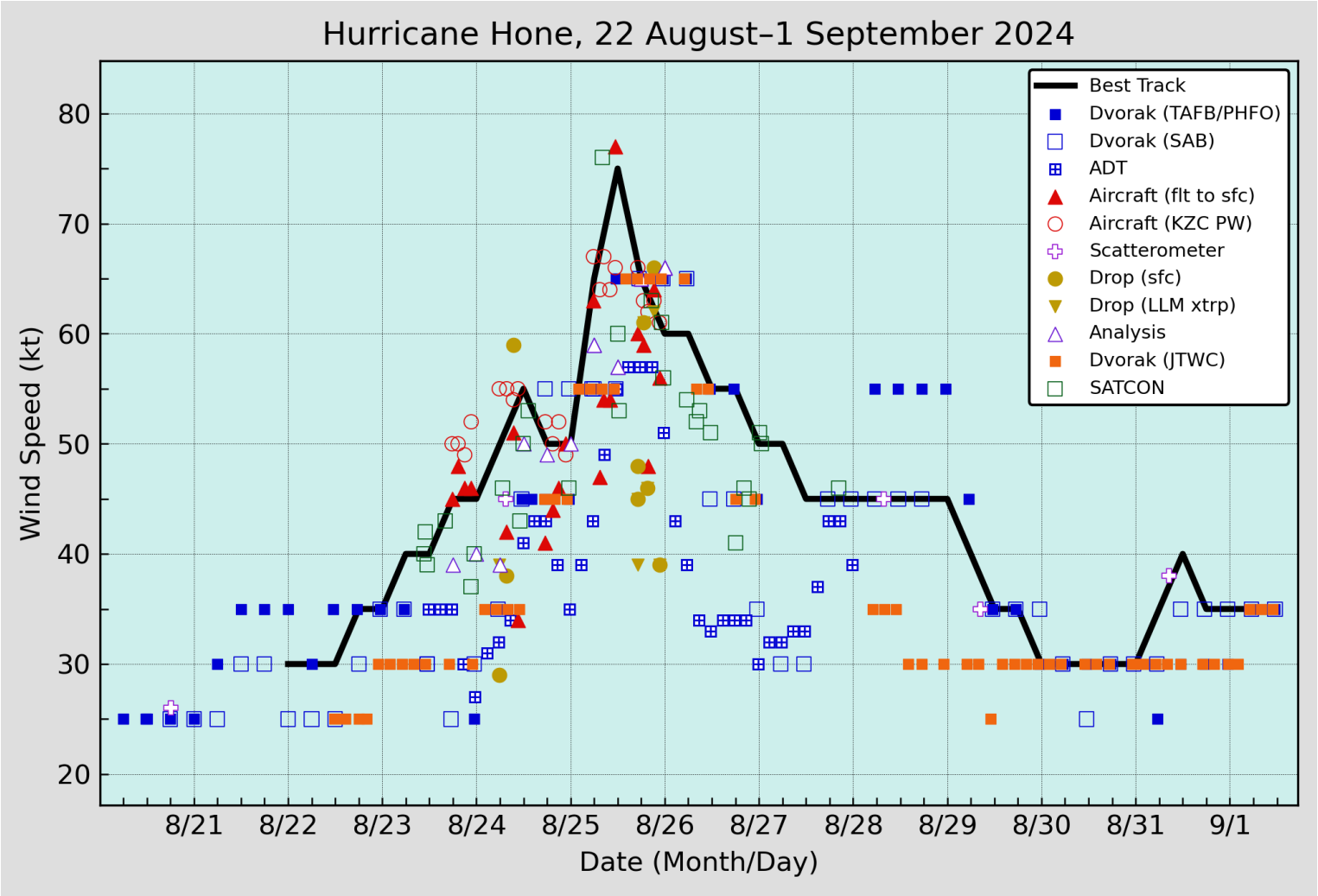


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Hone 22 August–1 September 2024. Aircraft observations have been adjusted for elevation using 90%, 80%, and 75% adjustment factors for observations from 700 mb, 850 mb, and 925 mb, respectively. Dropwindsonde observations include actual 10 m winds (sfc), as well as surface estimates derived from the mean wind over the lowest 150 m of the wind sounding (LLM). Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies.

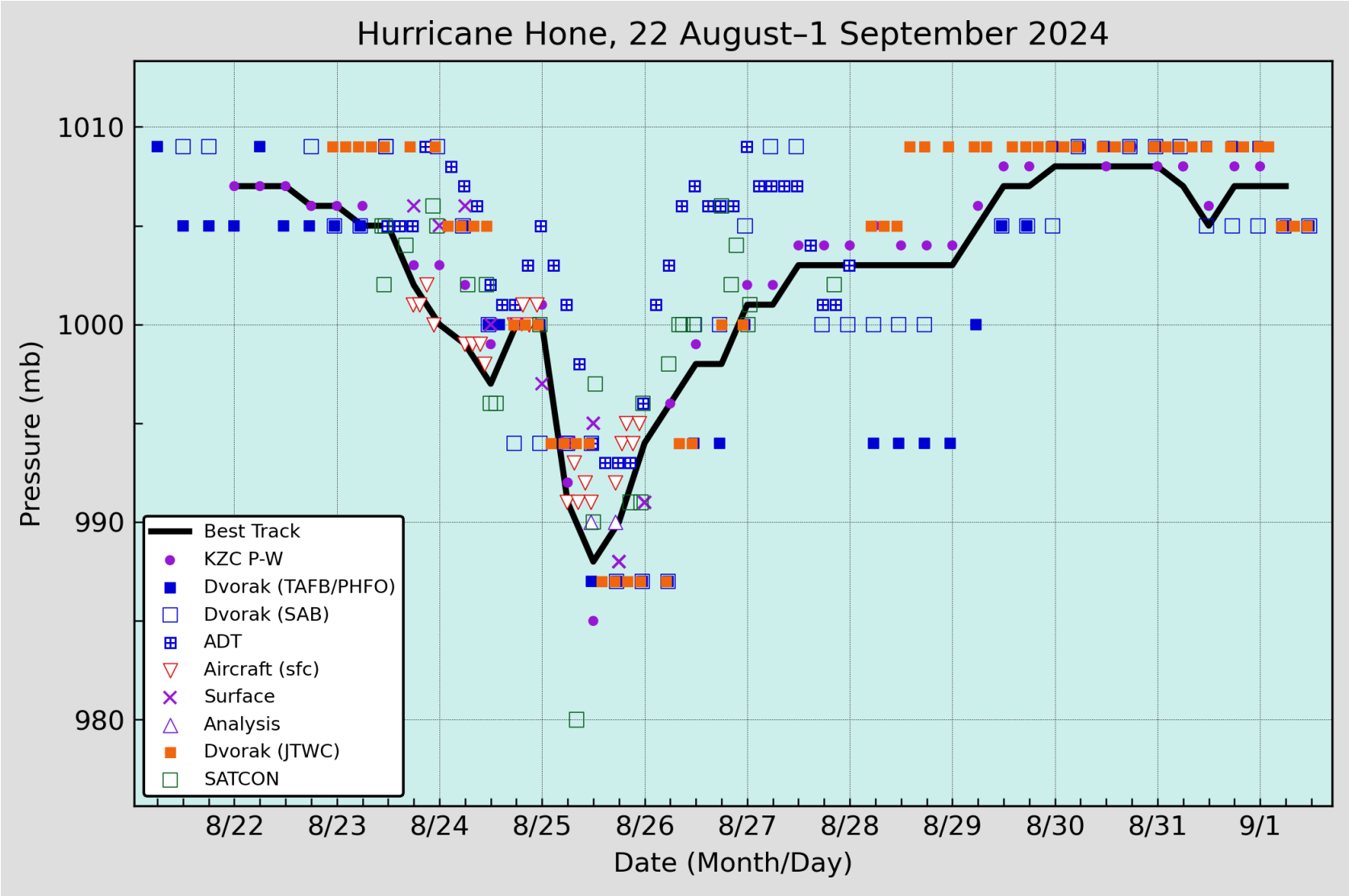


Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Hone 22 August–1 September 2024. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship.

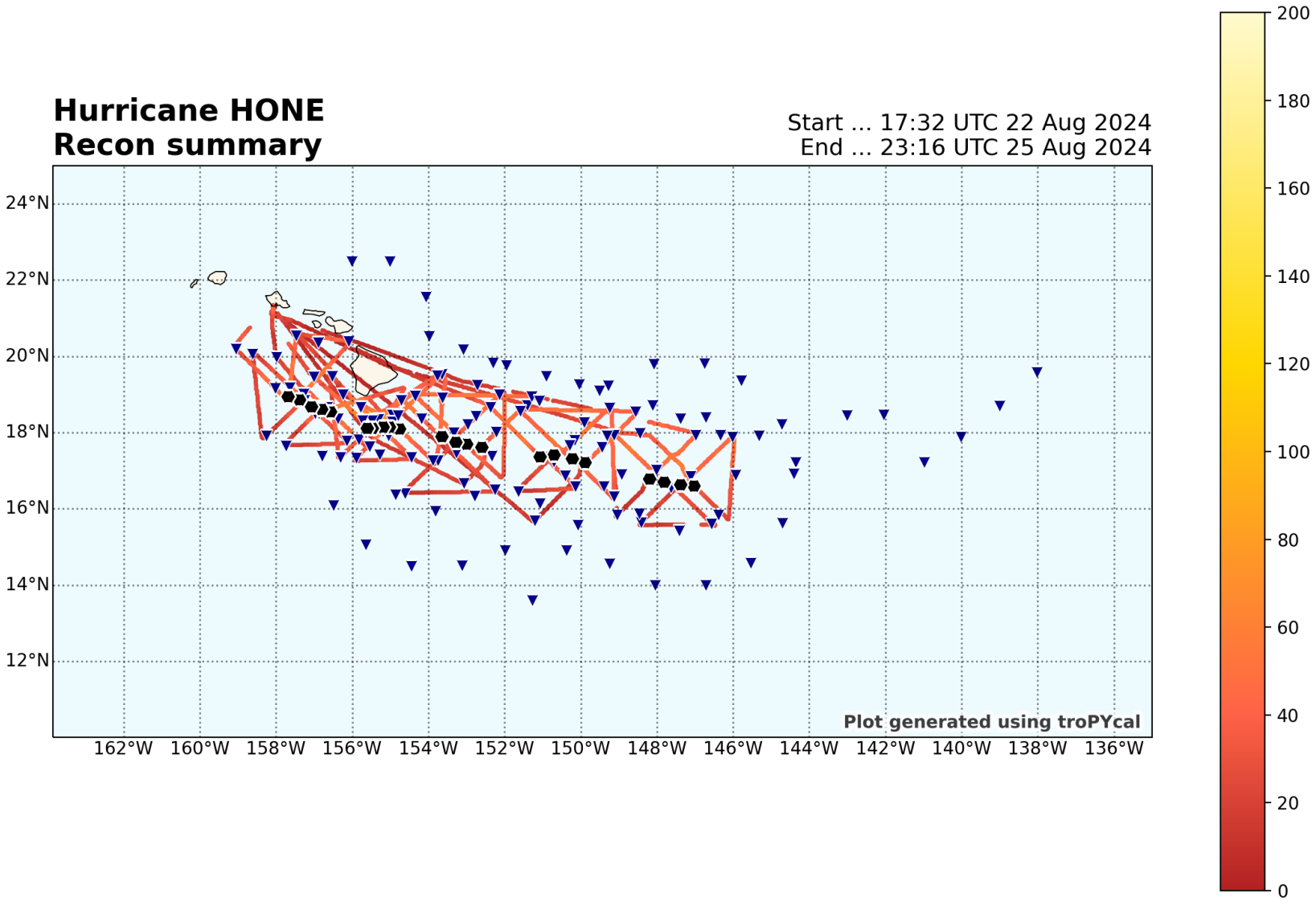


Figure 4. Air Force Reserve aircraft flight tracks (colored lines) from reconnaissance missions into Hurricane Hone from 22–25 August, 2024. The blue triangles indicate dropsonde locations. The color of the flight track represents the observed flight-level wind speed in knots at that location (see legend).

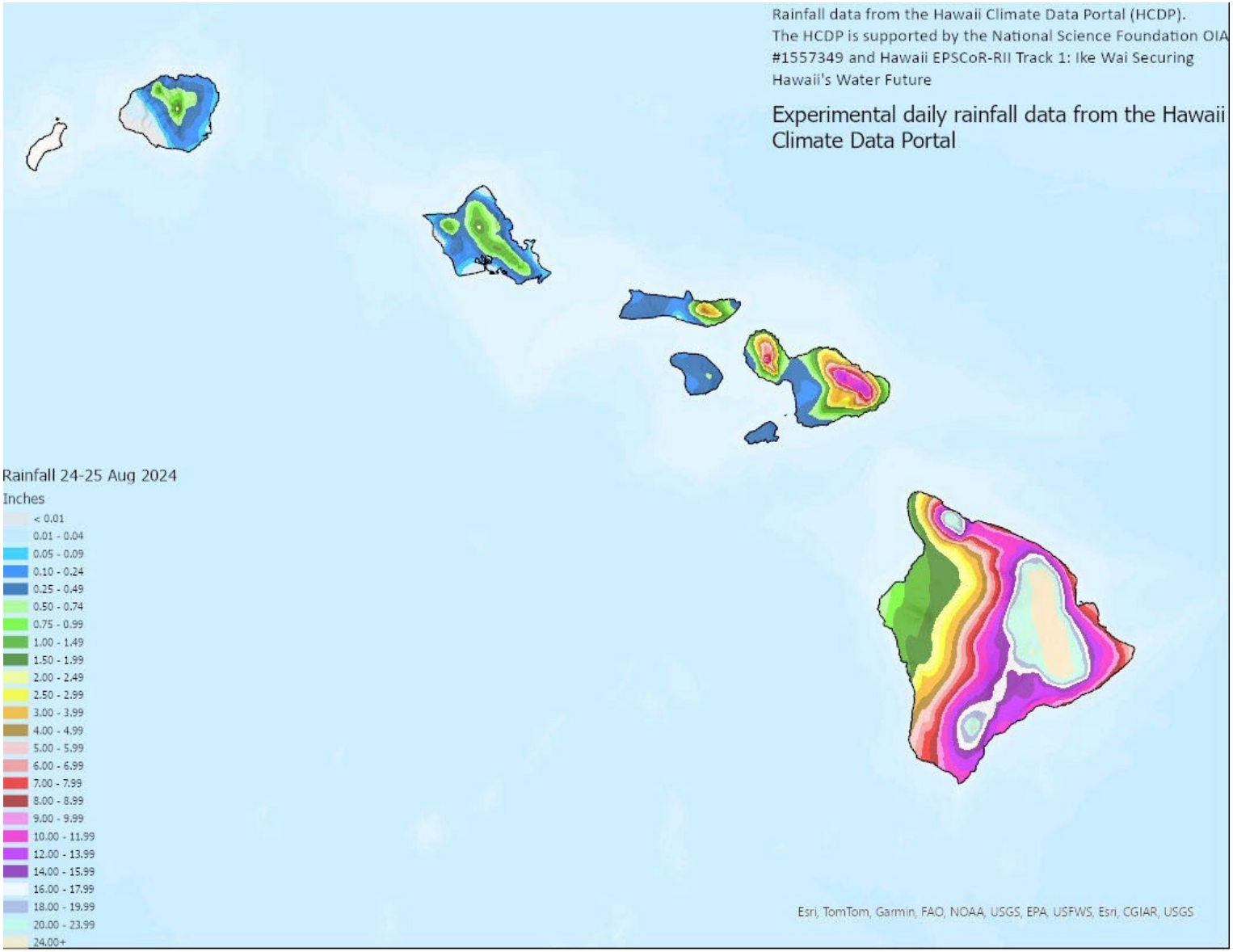


Figure 5. Rainfall values (inches) associated with Hurricane Hone, 24–25 August 2024. Image Courtesy of: Hawaii Climate Data Portal (https://www.hawaii.edu/climate-data-portal/).



Figure 6. Flash flood damage to the former “Hobbit House” bed-and-breakfast in Waiohinu, Hawaii associated with Hurricane Hone, 24–25 August 2024. Image Courtesy of: Bob Martin

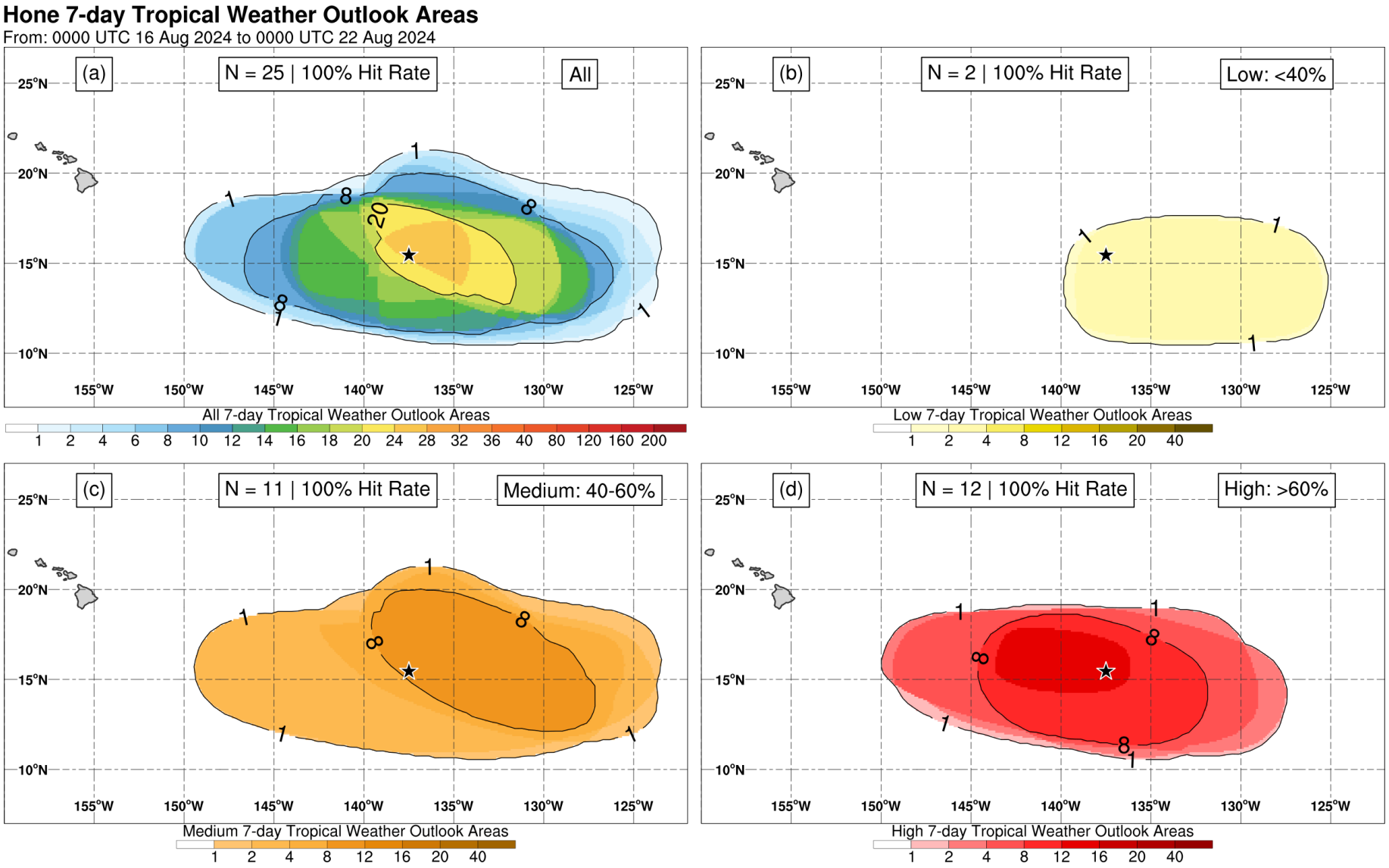


Figure 7. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Hone for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.